

SCSD2613

System Analysis and Design



TOPIC VI

Systems Implementation

Part Four: Training and Support, Organizational Issues

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update: August 2019 (sharinhh)

■ MAJOR TOPICS

SYSTEM IMPLEMENTATION

- System implementation process

SYSTEM CONSTRUCTION (CODING) & TESTING

- System construction
- Types of system testing

INSTALLATION

- Types of system installation
- Planning the installation

DOCUMENTING THE SYSTEM

- Generic guidelines

TRAINING AND SUPPORT

- Types of training
 - Support
- 

ORGANIZATIONAL ISSUES

- Factors
 - Security issues
 - Project closure
- 

■ MAJOR TOPICS

TRAINING AND SUPPORT

- Types of training
- Support



■ TRAINING AND SUPPORTING USERS

- **Training users**

- Potential training topics

1. Use of the system
2. General computer concepts
3. Information system concepts
4. Organizational concepts
5. System management
6. System installation

- **Support**

- providing ongoing educational and problem-solving assistance to information system users
 - For in-house developed systems, support materials and jobs will have to be prepared or designed as part of the implementation process.



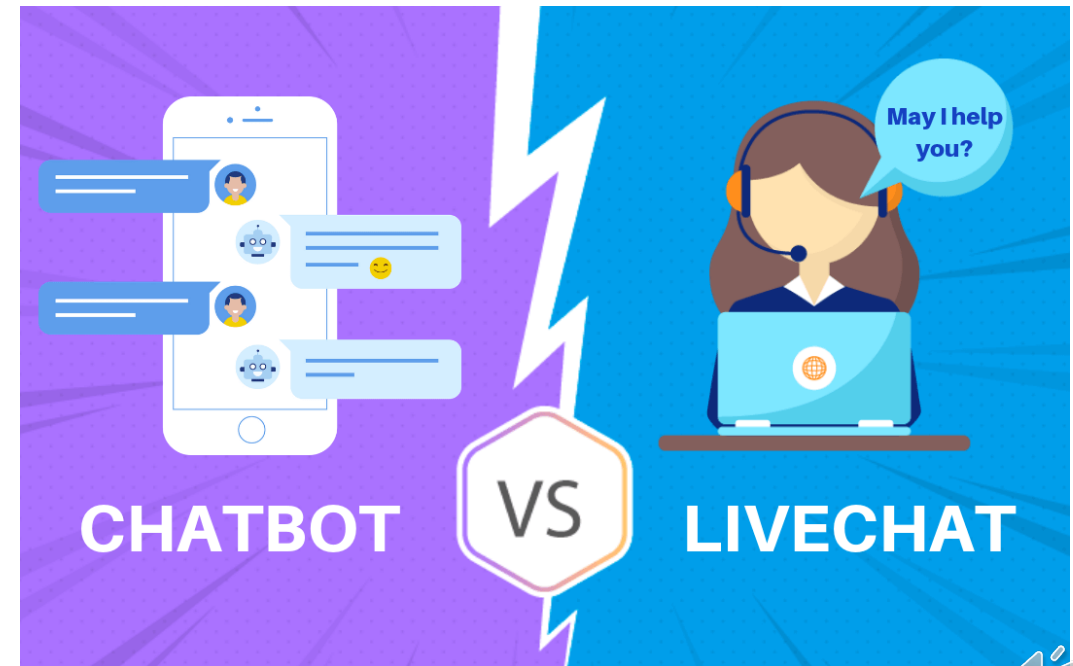
■ TYPES OF TRAINING METHODS

- Resident expert
- Traditional instructor-led classroom training
- E-learning, distance learning
- Blended learning (instructor plus e-learning)
- Software help components
- External sources (e.g. vendors)



■ SUPPORTING USERS

- Support is important to users, but has often been inadequate.
- Providing support can be expensive and time-consuming.
- Vendors usually charge for their support, using 900- numbers, or charge a fee for unlimited or monthly support.
- One approach is through **automation**.
 - Internet-based online support forums
 - On-demand fax
 - Voice response systems
 - Knowledge bases



■ PROVIDING SUPPORT THROUGH HELPDESK

- **Help desk:** a single point of contact for all user inquiries and problems about a particular information system or for all users in a particular department
- Requires
 - *Technical skills:* extensive knowledge about how to use the system and typical problems that can be encountered
 - *People skills:* good listening and communication, dealing with complaints and frustrations



■ SUPPORT ISSUES TO BE CONSIDER

- User questions and problems
- Recovery and backup
- Disaster recovery
- PC maintenance
- Writing newsletters
- Setting up user groups



■ MAJOR TOPICS

ORGANIZATIONAL ISSUES

- Factors
- Security issues
- Project closure



ORGANIZATIONAL ISSUES IN SYSTEM IMPLEMENTATION

- Why does implementation sometimes fail?
- Traditional wisdom of primary success factors:
 - Management support
 - User involvement
 - But these are not enough
- Other important factors
 - Commitment to project
 - Commitment to change
 - Extent of project definition and planning



■ FACTORS INFLUENCING SYSTEM USE

- Personal stake of users
- System characteristics
- User demographics
- Organizational support
- Performance
- Satisfaction

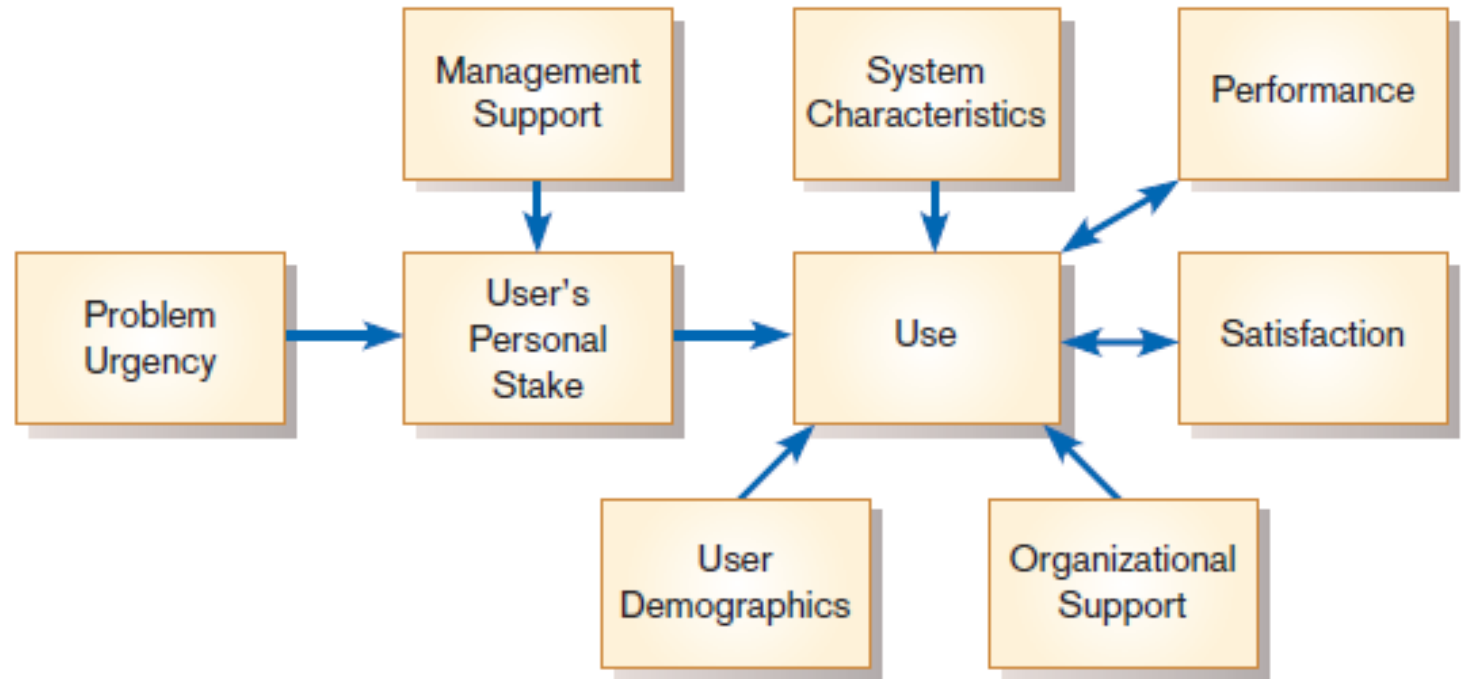


Figure 13-10

Implementation success

(Source: Adapted from Lucas, H. C. 1997. *Information Technology for Management*. New York: McGraw-Hill, with the permission of the McGraw-Hill Companies. All rights reserved.)

■ SECURITY ISSUES

- Increasingly important issue for organizations and their management
- **Malicious software** (*malware*): includes Trojan horses, worms, viruses, and other kinds
- External sources of threats include laptop theft, system penetration, and denial of service.

Nature of Breaches	
Malware infection	67%
Being fraudulently represented as sender of phishing messages	39%
Laptop/mobile theft	34%
Bots/zombies within the organization	29%
Insider abuse of internet access or e-mail	25%
Denial of service	17%
Unauthorized access or privilege escalation by insider	13%

Proportion of IT Budget Devoted to Security	
Proportion spending more than 10%	19%
Proportion spending between 2% and 10%	40%
Proportion spending between 1% and 2%	16%
Proportion spending less than 1%	10%

Security Technologies Used	
Antivirus software	97%
Firewalls	95%
Anti-spyware software	85%
Virtual private network	79%
Vulnerability/Patch management	68%
Encryption of data in transit	66%
Intrusion detection	62%

(Source: Data from Computer Security Institute [Richardson], 2011.)

PROJECT CLOSE-DOWN

- Evaluate team.
 - Reassign members to other projects.
- Notify all affected parties that the development project is ending and that you are switching to operation and maintenance mode.
- Conduct post project reviews.
- Close out customer contract.
 - Formal signoff



■ SUMMARY

- In this chapter you learned how to:
 - ✓ Describe the process of coding, testing, and converting an organizational information system and outline the deliverables and outcomes of the process.
 - ✓ Apply four installation strategies: direct, parallel, single-location, and phased installation.
 - ✓ List the deliverables for documenting the system and for training and supporting users.
 - ✓ Explain why system implementation sometimes fails.
 - ✓ Describe the threats to system security and remedies that can be applied.



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Thank You

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